AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

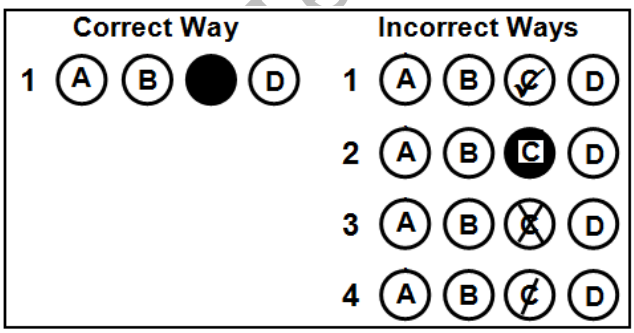
Chemistry Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Consider the given electronic configuration of an element **X**.

$$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}$$

The element **X** belongs to

	Period	Group	Block of Periodic Table
A	IV	IIB	d
B	IV	VIA	p
C	V	IIB	d
D	V	VIA	p

2. Which of the following elements has the HIGHEST melting point?

A. carbon.
B. lithium.
C. fluorine.
D. magnesium.

3. When magnesium chloride ($MgCl_2$) dissolves in water, it forms a/ an

A. basic solution (pH = approximately 9).
B. acidic solution (pH = approximately 6).
C. acidic solution (pH = approximately 3).
D. basic solution (pH = approximately 12).

4. Which of the following electronic configurations represents the ion of a transition element?

A. $Y^{1-} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^0$
B. $X^{1+} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^0$
C. $Y^{2-} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^0$
D. $X^{2+} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^1, 4s^0$

5. To avoid the oxidation of vanadium (II) ions in solution, it should be kept in the

A. absence of oxygen and zinc.
B. presence of oxygen and zinc.
C. presence of oxygen and absence of zinc.
D. absence of oxygen and presence of zinc.

6. When there is crowding within the substrate molecule, then elimination reaction is favoured over substitution reaction, because the removal of

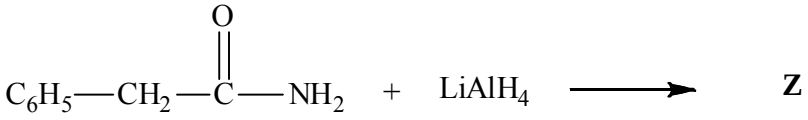
A. β-hydrogen atom by base from tertiary planar carbonium ion is easy.
B. α-hydrogen atom by base from tertiary planar carbonium ion is easy.
C. β-hydrogen atom by base from tertiary planar carbonium ion is difficult.
D. α-hydrogen atom by base from tertiary planar carbonium ion is difficult.

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7. All of the following are TRUE about S_N2 reactions EXCEPT they

   A. are favoured in non-polar solvent.
   B. always occur with partial racemisation.
   C. involve the formation of transition state.
   D. always occur with inversion of configuration.

8. The missing product **Z** in the given chemical reaction is



	Product Z
A	$C_6H_5—CH_2—NH_2$
B	$C_6H_5—CH_3$
C	$C_6H_5—CH_2—CH_2—NH_2$
D	$C_6H_5—CH_2—CH_3$

9. The anion formed by the deprotonation of alcohol is

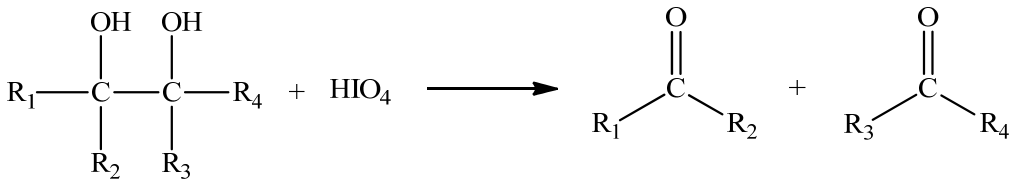
   A. carbanion.
   B. acetate ion.
   C. carbocation.
   D. alkoxide ion.

10. The CORRECT order of reactivity of alcohol with reference to O–H bond cleavage is

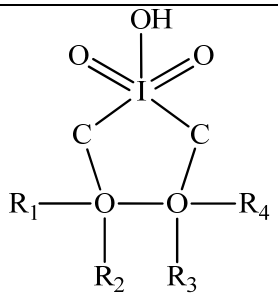
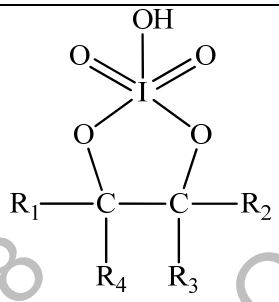
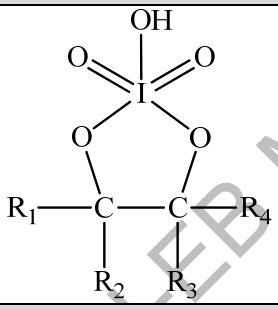
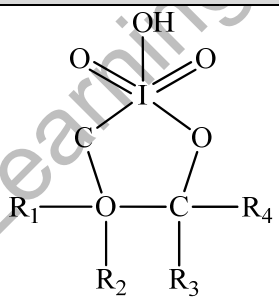
   A. primary alcohol > CH_3OH > secondary alcohol > tertiary alcohol.
   B. tertiary alcohol > secondary alcohol > primary alcohol > CH_3OH.
   C. tertiary alcohol > secondary alcohol > CH_3OH > primary alcohol.
   D. CH_3OH > primary alcohol > secondary alcohol > tertiary alcohol.

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11. The oxidative cleavage of 1,2-diol by periodic acid (HIO_4) gives two carbonyl compounds. The reaction occurs via the formation of a cyclic periodate ester.

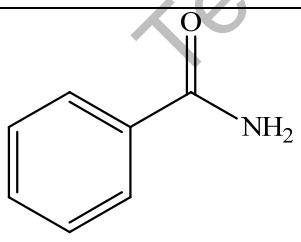
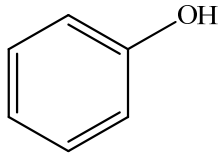
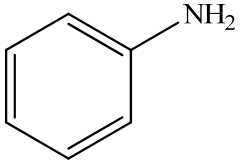
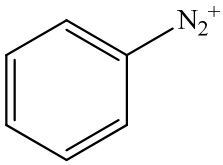


The structure of cyclic periodate intermediate formed in the given reaction would be

	
A	**B**
	
C	**D**

12. A diazonium salt forms when aromatic amine reacts with nitrous acid at low temperature.

The product obtained when benzene diazonium ion reacts with water is

	
A	**B**
	
C	**D**

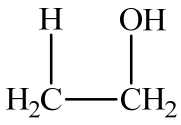
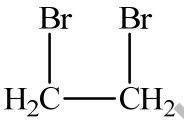
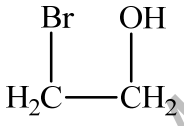
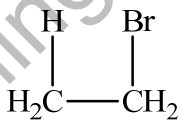
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13. Stereoisomerism is exhibited by compounds which have different arrangement of atoms in space.

A compound is optically active if it

A. can superimpose on its mirror image.
B. has atleast one asymmetric centre with plane of symmetry.
C. has atleast one asymmetric centre with no plane of symmetry.
D. contains carbon atom which is bonded to three different groups.

14. The product obtained when ethene is treated with bromine in the presence of water is

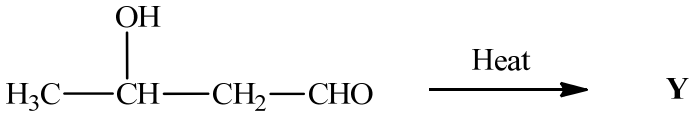
	
A	**B**
	
C	**D**

15. Which of the following CORRECTLY matches the common name of the organic compound with its IUPAC name?

	Common Name	IUPAC Name
A	Methyl propyl ketone	2-butanone
B	Dimethyl ketone	2-propanone
C	Methyl ethyl ketone	Propanone
D	Methyl isopropyl ketone	3-pentanone

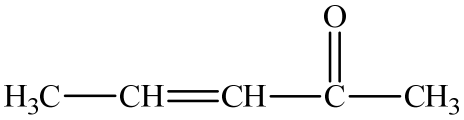
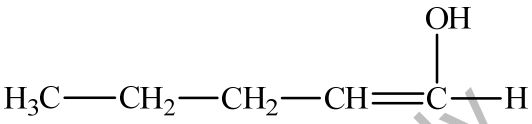
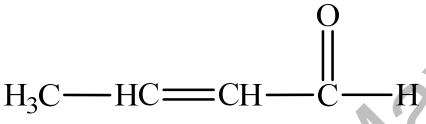
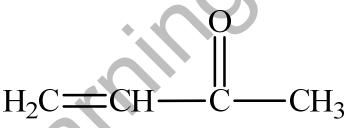
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16. When the aldol compound **X** is heated in the presence of dilute acid, it forms an unsaturated carbonyl compound **Y**.

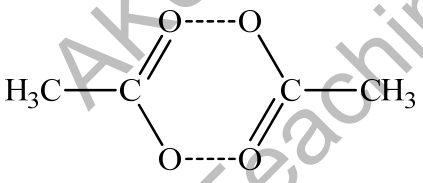
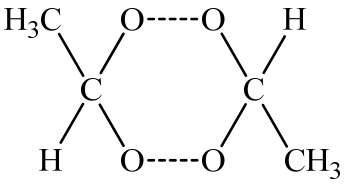
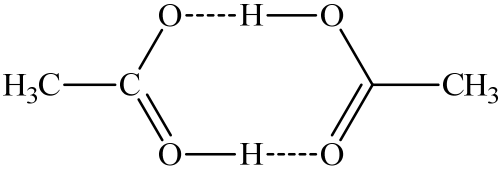
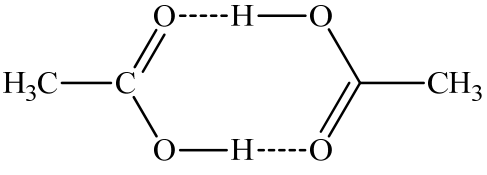


X

The structure of compound **Y** is

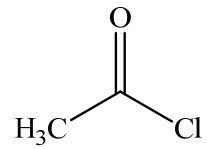
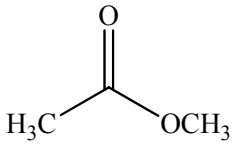
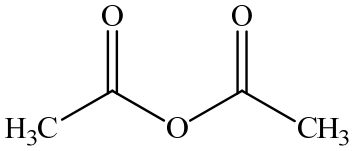
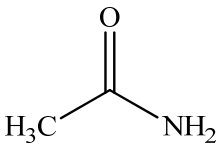
	
A	**B**
	
C	**D**

17. The structure that represents the dimer of carboxylic acid is

	
A	**B**
	
C	**D**

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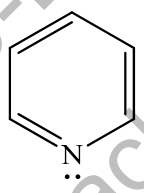
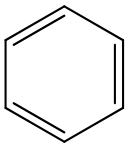
18. The carboxylic acid derivative which is the LEAST reactive is

	
A	**B**
	
C	**D**

19. The type of organic compound which is prepared by replacing active hydrogen of an organic acid with alkyl or aryl groups is

A. ester.
B. ether.
C. aldehyde.
D. acid halide.

20. The compound which is alicyclic in nature is

	
A	**B**
$CH_3—CH_2—CH_2—CH_3$	
C	**D**

21. $CH_3COOC_3H_7$ is the structural formula of

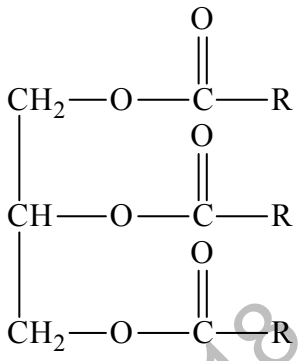
A. ethanoic acid.
B. propanoic acid.
C. ethyl ethanoate.
D. propyl ethanoate.

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22. The enzyme which helps in the digestion of starch in the human body is

A. lipase.
B. pepsin.
C. trypsin.
D. amylase.

23. The hydrolysis of given triglyceride molecule by lipase enzyme will give



Triglyceride

A. 1 molecule of alcohol and 3 molecules of fatty acid.
B. 1 molecule of alcohol and 2 molecules of fatty acid.
C. 2 molecules of alcohol and 1 molecule of fatty acid.
D. 3 molecules of alcohol and 3 molecules of fatty acid.

24. The dye that contains azo group in its structure is

A. indigo blue.
B. indigo white.
C. methyl orange.
D. phenolphthalein.

25. The compound used as plasticiser in nail polish composition is

A. pearl.
B. acetone.
C. castor oil.
D. nitrocellulose.

26. Hypochlorous acid is formed during the process of water purification. All of the following methods can be used to generate hypochlorous acid EXCEPT

A. adding small amount of HCl in water.
B. reacting sodium hypochlorite with water.
C. reacting calcium hypochlorite with water.
D. dissolving molecular chlorine gas in water.

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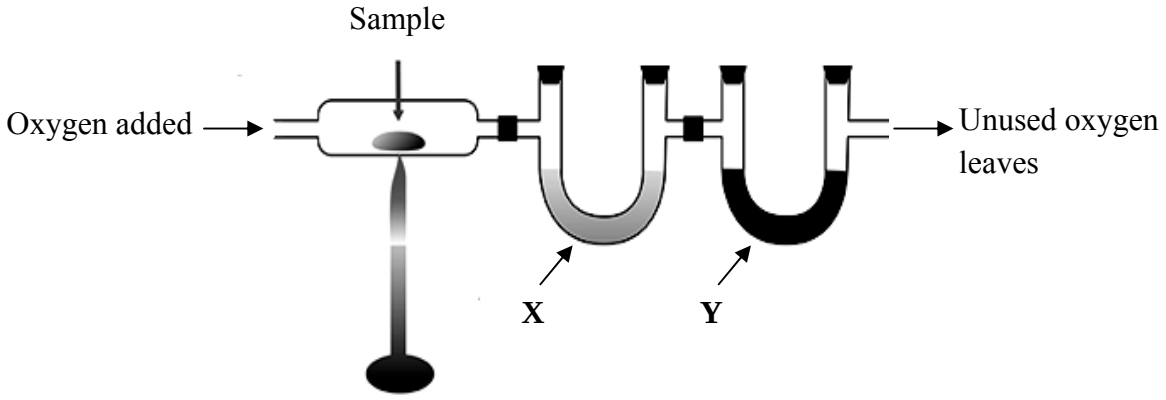
27. The compound which can be used as an alternate to chlorofluorocarbons in refrigeration is

A. methyl chloride.
B. difluoromethane.
C. carbon tetrachloride.
D. trichlorofluoromethane.

28. Ultraviolet-visible spectroscopy involves

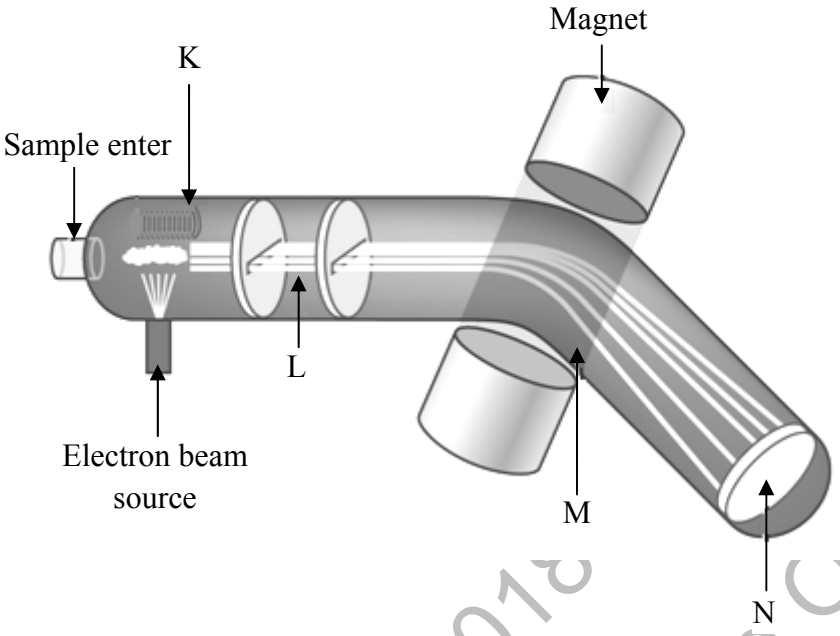
A. measurement of mass to charge ratio of ions.
B. bombardment of molecules with high energy electrons.
C. absorption of radiation by compounds with conjugated double bonds.
D. determination of number of different types of protons in the molecule.

29. The parts, **X** and **Y**, of the apparatus in the given diagram of combustion analysis can be identified as



	X	Y
A	Carbon dioxide Absorber	Water Absorber
B	Water Absorber	Hydrogen Absorber
C	Hydrogen Absorber	Carbon dioxide Absorber
D	Water Absorber	Carbon dioxide Absorber

30. A mass spectrometre ionises molecules into ions and separates them based on their mass-to-charge ratio, which allows to determine the mass of the molecule.



In the given diagram of mass spectrometre, the part that involves the acceleration of ions is

A. K.
B. L.
C. M.
D. N.

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